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NANOFLUID OVER AN EXPONENTIALLY STRETCHING PERMEABLE SHEET WITH VISCOUS DISSIPATION

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Abstract: *The present study analyzes the steady boundary layer flow of magneto-nanofluid due to an exponentially permeable stretching sheet with viscous dissipation. In this paper, the effect viscous dissipation on heat transfer and nanoparticle volume friction is considered. The similarity ordinary differential equations were then solved by MATLAB **bvp4c** solver. The study reveals that the governing parameters, namely, the magnetic parameter, wall mass suction parameter, Prandtl number, the Lewis number, Eckert number, Brownian motion parameter, and thermophoresis parameter, have major effects on the flow field, the heat transfer, and the nanoparticle volume fraction as well as skin friction, local Nusselt number and local Sherwood number has been discussed in detail.*

Keywords: *MHD, nanoparticle, heat transfer, viscous dissipation, stretching sheet.*

1. Introduction

Choi [1] was the first person to introduce the word "nanofluids. Heat transfer phenomena over a stretching sheet have received great attention in chemical and manufacturing processes. The pioneering work of Sakiadis[2], Crane [3] studied a steady flow past a stretching sheet and presented a closed form solution to it.

Suction or injection (blowing) of a fluid through the bounding surface can significantly change the flow field. Recently the effects of variable suction and thermophoresis on steady MHD flow over a permeable inclined plate was analyzed by Alam et al. [4] while the heat and mass transfer of thermophoretic hydromagnetic flow with lateral mass flux, heat source.

Liu [5] investigated the flow and heat transfer of an electrically conducting fluid of second grade over a stretching sheet subject to a transverse magnetic field. Koo and Kleinstreuer [6] have investigated the effects of viscous dissipation on the temperature field using dimensional analysis and experimentally validated computer simulations. Yazdi et al. [7] evaluates the slip MHD flow and heat transfer of an electrically conducting liquid over a permeable surface in the presence of the viscous dissipation effects under convective boundary conditions.

2. MATHEMATICAL FORMULATION

Consider the steady boundary layer flow of nanofluid over an exponentially stretching sheet in presence of a transverse magnetic field. The governing equations of motion and the energy equation may be written in usual notation as [8,9,10]

$$\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}=0 \tag{2.1}$$

$$u\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}=\nu\frac{\partial^2 u}{\partial y^2}-\frac{\sigma B^2}{\rho_f}u \tag{2.2}$$

$$u\frac{\partial T}{\partial x}+v\frac{\partial T}{\partial y}=\alpha\frac{\partial^2 T}{\partial y^2}+\frac{(\rho c)_p}{(\rho c)_f}\left[D_B\frac{\partial N}{\partial y}\frac{\partial T}{\partial y}+\frac{D_T}{T_\infty}\left(\frac{\partial T}{\partial y}\right)^2\right]+\frac{\mu_f}{(\rho c)_f}\left(\frac{\partial u}{\partial y}\right)^2 \tag{2.3}$$